

StakeDXF

User Guide

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Trimble TSC5 field controller app

TRIO Engineering

1. Installation (Trimble TSC5)

StakeDXF ships as an Android APK for the Trimble TSC5.

File: dist/StakeDXF-tsc5.apk (~65 MB)

Package: com.stakedxf.stakedxf

Steps

1. Copy StakeDXF-tsc5.apk onto the TSC5 (USB File Transfer, or any file share you already use).
2. On the TSC5 open Files / Downloads and tap the APK.
3. If Android blocks it, enable Allow from this source for Files (or the app that opened the APK).
4. Tap Install, then open StakeDXF from the app drawer.

If Install is greyed out

Company MDM may block unknown APKs. Ask IT to allow unknown-source installs or push the APK through managed distribution.

Common mistakes

- ? Trying to open the APK on iPhone ? iOS cannot install Android APKs.
- ? Opening a zip/repo instead of the .apk file.
- ? USB mode left on Charging only ? switch to File Transfer.

2. Usage · Convert DWG · DXF

Purpose: turn a Civil 3D drawing into a Trimble-stakeable DXF on the controller.

Steps

1. Open StakeDXF ? Convert DWG ? DXF.
2. Tap Choose DWG / DXF and pick the office drawing.
3. Tap Convert for Trimble Access.
4. Confirm stakeable entity count (and proxy explode count when Civil 3D proxies were present).
5. Review Converted layers (empty layers are omitted).
6. Check layers to include, then Save DXF (or share the full DXF).
7. Place it in:
Trimble Data/Projects/<your project>/
6. In Trimble Access: Map ? Layer manager ? Map files ? make the DXF selectable ? Stakeout.

What happens on-device

- ? LibreDWG reads the DWG into DXF.
- ? ezdxf explodes ACAD_PROXY_ENTITY / AEC proxy graphics into LINE / ARC / POLYLINE.
- ? Non-stakeable entity types are filtered out.

Tip: Office DWGs should be saved with proxy graphics preserved so Civil 3D features can be recovered.

3. Usage · Export Points & staking plots

Purpose: build a scaled field sheet from selected points (and optional DXF linework).

Steps

1. In Trimble Access, export the points you need as CSV / TXT (PNEZD).
2. StakeDXF ? Export Points.
3. Import points CSV / TXT.
4. (Optional) Link DXF linework and choose layers to draw.
5. Set Point marker and Point label format.
6. Leave Include point list table off unless you want the coordinate table.
7. (Optional) Add from object library ? place, nudge, scale, rotate, recolor.
8. Select the points for the sheet.
9. Tap Create staking plot PDF ? scale is chosen automatically to fit.
10. Or tap Export selected points CSV for a trimmed point list.

Supported point formats

- ? Headered or headerless PNEZD: Point, Northing, Easting, Elevation, Description
- ? Also accepts common header aliases (Point #, Elev, Desc, etc.)

Linked DXF linework

- ? Reads LINE, LWPOLYLINE, POLYLINE, ARC, CIRCLE
- ? Layer checklist lets you keep only relevant linework on the plot

4. Plot customization reference

Markers

Filled triangle · Triangle outline · Cross (+) · X · Large X · Circle · Dot · Large dot

Labels

Point number · Number + description · Number + elevation ·

Number + description + elevation · No labels

Layout

? Point list table OFF by default ? gives the plan view more sheet space.

? Turn the table ON for a control-note style coordinate list.

? Sheet size is ANSI B landscape (17" × 11"), north up, auto engineering scale.

Examples

See [dist/plot_examples/](#) for ten sample PDFs covering these combinations.

Regenerate with: `cd mobile/stakedxf && dart run tool/generate_plot_examples.dart`

5. Help & troubleshooting

No stakeable entities after convert

- ? Confirm the DWG still contains proxy graphics (ACAD_PROXY_ENTITY).
- ? Civil objects saved without proxies cannot be exploded to linework.
- ? Try converting a DXF that already shows proxies in a CAD viewer.

Staking plot has no linework

- ? Link a DXF and enable Draw linked DXF linework.
- ? Ensure at least one layer is checked.
- ? Only LINE / LWPOLYLINE / POLYLINE / ARC / CIRCLE are drawn.

Points missing after import

- ? Use PNEZD order or a headered CSV with Northing/Easting columns.
- ? Blank or malformed rows are skipped.

App will not install

- ? Must be the TSC5 (Android), not iPhone.
- ? Enable Allow from this source.
- ? MDM may require IT approval.

Support files

- ? dist/INSTALL_TSC5.md
- ? dist/StakeDXF_UI_Slide_Deck.pdf
- ? docs/USER_GUIDE.md
- ? GitHub PR for StakeDXF TSC5 app

6. Quick reference card

Home

Convert DWG ? DXF Recover Civil 3D linework
Export Points CSV + staking plot PDF

Convert

Choose drawing ? Convert ? Save DXF ? Trimble Map files

Export Points

Import CSV ? (Link DXF) ? Marker/Label options ? Select points
? Create staking plot PDF or Export CSV

Good defaults for staking

Marker: Large X
Label: Number + elevation
Point list table: Off
Linework: On (only layers you need)

Runs on-device. No cloud upload.